Laboratory Assignment File

for

Cryptography Laboratory (CS0557)

Master of Technology

in

Computer Science & Engineering

Submitted by

Yugal

(Roll no. 25903053)

Submitted to

Dr. Harsh Kumar Verma

(Assistant Professor, NIT Jalandhar)

Department of Computer Science & Engineering

Dr. B R Ambedkar National Institute of Technology Jalandhar

Punjab, India-144008

May, 2026

Contents

Assignment 1: Basic Programs **28-01-2026** **1**

1.1 Write a program to print prime numbers between 1 and 1000. 1

1.2 Write a program to perform matrix multiplication. 1

1.3 Write a program to find inverse of a matrix. 4

List of Figures

1 Program to print prime numbers between 1 and 1000 1
2 Program to perform matrix multiplication 4
3 Program to find inverse of a matrix 6

Assignment 1: Basic Programs

1.1 Write a program to print prime numbers between 1 and 1000.

```java
import java.util.ArrayList;
import java.util.List;

public class Prime {
    public static void main(String[] args) {
        List<Integer> primes = new ArrayList<>();
        int x = 1;
        int y = 100;

        for (int i = 2; i <= y; i++) {
            boolean isPrime = true;

            for (int j = 2; j <= i / 2; j++) {
                if (i % j == 0) {
                    isPrime = false;
                    break;
                }
            }

            if (isPrime) {
                primes.add(i);
            }
        }

        System.out.println("Prime numbers between " + x + " and " + y + ":");
        System.out.println(primes);
    }
}
```

Output

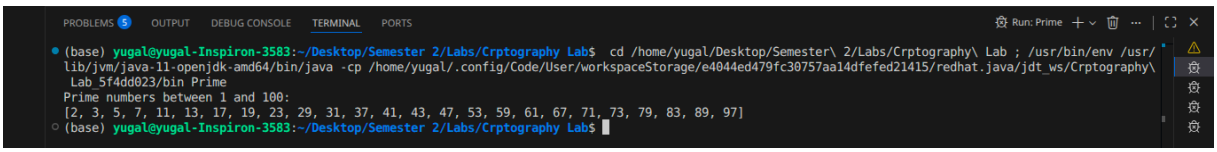

Figure 1: Program to print prime numbers between 1 and 1000

1.2 Write a program to perform matrix multiplication.

```java
import java.util.Scanner;

public class Matrix_Multiplication {

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("Enter the number of rows for the first matrix (A):");
        int rowsA = scanner.nextInt();
        System.out.println("Enter the number of columns for the first matrix (A):");
        int colsA = scanner.nextInt();

        System.out.println("Enter the number of rows for the second matrix (B):");
        int rowsB = scanner.nextInt();
        System.out.println("Enter the number of columns for the second matrix (B):");
        int colsB = scanner.nextInt();

        if (colsA != rowsB) {
            System.out.println("\nError: Matrix multiplication is not possible.");
            System.out.println("The number of columns in the first matrix must equal the number of rows in the second matrix.");
            return;
        }

        int[][] matrixA = new int[rowsA][colsA];
        int[][] matrixB = new int[rowsB][colsB];
        int[][] resultMatrix = new int[rowsA][colsB];

        System.out.println("\nEnter elements of the first matrix (A):");
        for (int i = 0; i < rowsA; i++) {
            for (int j = 0; j < colsA; j++) {
                System.out.print("A[" + i + "][" + j + "]: ");
                matrixA[i][j] = scanner.nextInt();
            }
        }

```

```java
        System.out.println("\nEnter elements of the second matrix (B):");
        for (int i = 0; i < rowsB; i++) {
            for (int j = 0; j < colsB; j++) {
                System.out.print("B[" + i + "][" + j + "]: ");
                matrixB[i][j] = scanner.nextInt();
            }
        }

        for (int i = 0; i < rowsA; i++) {
            for (int j = 0; j < colsB; j++) {
                for (int k = 0; k < colsA; k++) {
                    resultMatrix[i][j] += matrixA[i][k] * matrixB[k][j];
                }
            }
        }

        System.out.println("\nResultant Matrix:");
        for (int i = 0; i < rowsA; i++) {
            for (int j = 0; j < colsB; j++) {
                System.out.print(resultMatrix[i][j] + " ");
            }
            System.out.println();
        }

        scanner.close();
    }
}
```

Output

```
(base) yugal@yugal-Inspiron-3583:~/Desktop/Semester 2/Labs/Crptography Lab$  cd /home/yugal/Desktop/Semester\ 2/Labs/Crptography\
 Lab ; /usr/bin/env /usr/lib/jvm/java-11-openjdk-amd64/bin/java -cp /home/yugal/.config/Code/User/workspaceStorage/e4044ed479fc30
757aa14dfefed21415/redhat.java/jdt_ws/Crptography\ Lab_5f4dd023/bin Matrix_Multiplication
Enter the number of rows for the first matrix (A):
3
Enter the number of columns for the first matrix (A):
3
Enter the number of rows for the second matrix (B):
3
Enter the number of columns for the second matrix (B):
3

Enter elements of the first matrix (A):
A[0][0]: 1
A[0][1]: 2
A[0][2]: 3
A[1][0]: 4
A[1][1]: 5
A[1][2]: 6
A[2][0]: 7
A[2][1]: 8
A[2][2]: 9

Enter elements of the second matrix (B):
B[0][0]: 1
B[0][1]: 2
B[0][2]: 3
B[1][0]: 4
B[1][1]: 5
B[1][2]: 6
B[2][0]: 7
B[2][1]: 8
B[2][2]: 9

Resultant Matrix:
30 36 42
66 81 96
102 126 150
(base) yugal@yugal-Inspiron-3583:~/Desktop/Semester 2/Labs/Crptography Lab$ ^C
```

Figure 2: Program to perform matrix multiplication

1.3 Write a program to find inverse of a matrix.

```java

import java.util.Scanner;

public class MatrixInverseSimple {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        double[][] a = new double[3][3];

        System.out.println("Enter elements of 3x3 matrix:");

        // Input
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                a[i][j] = sc.nextDouble();
            }
```

```java
        }

        // Determinant
        double det =
                a[0][0] * (a[1][1]*a[2][2] - a[1][2]*a[2][1])
              - a[0][1] * (a[1][0]*a[2][2] - a[1][2]*a[2][0])
              + a[0][2] * (a[1][0]*a[2][1] - a[1][1]*a[2][0]);

        if (det == 0) {
            System.out.println("Matrix is singular. No inverse.");
            return;
        }

        // Adjoint matrix
        double[][] adj = new double[3][3];

        adj[0][0] =  (a[1][1]*a[2][2] - a[1][2]*a[2][1]);
        adj[0][1] = -(a[0][1]*a[2][2] - a[0][2]*a[2][1]);
        adj[0][2] =  (a[0][1]*a[1][2] - a[0][2]*a[1][1]);

        adj[1][0] = -(a[1][0]*a[2][2] - a[1][2]*a[2][0]);
        adj[1][1] =  (a[0][0]*a[2][2] - a[0][2]*a[2][0]);
        adj[1][2] = -(a[0][0]*a[1][2] - a[0][2]*a[1][0]);

        adj[2][0] =  (a[1][0]*a[2][1] - a[1][1]*a[2][0]);
        adj[2][1] = -(a[0][0]*a[2][1] - a[0][1]*a[2][0]);
        adj[2][2] =  (a[0][0]*a[1][1] - a[0][1]*a[1][0]);

        // Inverse = adjoint / determinant
        System.out.println("Inverse Matrix:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.printf("%6.2f ", adj[i][j] / det);
            }
            System.out.println();
        }

        sc.close();
    }
}
```

Output

```
(base) yugal@yugal-Inspiron-3583:~/Desktop/Semester 2/Labs/Crptography Lab$  cd /home/yugal/Desktop/Semester\ 2/Labs/Crptography\
 Lab ; /usr/bin/env /usr/lib/jvm/java-11-openjdk-amd64/bin/java -cp /home/yugal/.config/Code/User/workspaceStorage/e4044ed479fc30
757aa14dfefed21415/redhat.java/jdt_ws/Crptography\ Lab_5f4dd023/bin MatrixInverseSimple
Enter elements of 3x3 matrix:
2 3 5
6 7 3
4 5 9
Inverse Matrix:
 -2.40   0.10   1.30
  2.10   0.10  -1.20
 -0.10  -0.10   0.20
```

Figure 3: Program to find inverse of a matrix